
Ravneet Singh

Tech Lead / Project Manager (<https://techsingh.wixsite.com/website>) (<https://github.com/rssinghji>)
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PROFESSIONAL SUMMARY

More than 13 years of extensive experience in developing, debugging, managing and maintaining backend, middleware, firmware, including both embedded as well as non-embedded systems and UI applications.
7+ years of experience in backend development for web apps using goLang.
6+ years of leading and managing teams for software and saas products.
Experience in software quality assurance and inspections.
Background with risk management, requirements reviews, software metrics, quality as well as test plans, test designs/cases; change control, and configuration management.
Proven ability and experience to program in C, C++ for embedded microprocessors, middleware and UI applications. Background developing in both Linux, MAC and Windows environments.
Experience programming in HTML, CSS and JavaScript, GoLang/Beego for web applications.
Experience programming in Python for middleware and testing applications.
Working knowledge of React.
Fast learner with previous work experience under demanding, problem-oriented circumstances.
Self-motivated and energetic team player.
Conducted guest lectures, leadership sessions and technical trainings.
Experience in Automation testing, through TDD.

Strengths:

Commitment, willpower and leadership.
Adaptable to environmental changes and emerging trends.
Willingness to learn and adaptability to put learning into practice.
Added a lot of innovative ideas for solutions and intellectual properties.

EDUCATION

- Master's in Business Administration with focus on Information Systems Management at University of Texas at Arlington (2021-2023) (GPA-4.0)
 - President (and founder) of MBA Student Association
 - Member of UTA Leadership Honors Program for graduate students
 - Member of Beta Alpha Psi at UTA
 - Member of Cyber Security Org at UTA
 - Member of Cultural Engagement and Social Change org
 - Member by invitation into BetaGammaSigma and PhiKappaPhi
- Masters in Electrical & Computer Engineering at Illinois Tech -10 W 35th St, Chicago, IL 60616, USA. From 22nd Aug 2016 to 13th May 2018.
- M.S from Manipal University, India - Tiger Circle Road, Madhav Nagar, Manipal, Karnataka 576104, India From 6th June 2011 to 8th July 2013 in Digital Design & Embedded Systems.
- B.TECH in Electronics & Communication Engineering from JNTU, India - Sir Mokshagundam Visvesvaraya Road, Saradha Nagar, Anantapur, Andhra Pradesh 515002, India from 4th September 2006 to 16th August 2010.

PROJECT TITLES

(Undergraduate)

Emergency Light
Industrial Training In Communications (AAI)
Fire Extinguishing Robot

(Graduate)

Chat machine using MODBUS protocol
Logic compilation using C++
Leadership Legacy Project

TECHNICAL SKILLS

Operating Systems	DOS, Windows98/XP/7, Linux, MAC
Languages	GoLang, C, C++, Python, Android(Java), JavaScript(basic), React, REST APIs, HTML, XML, CSS, JSON
Technologies	Beego Framework, MODBUS(TCP/RTU), ZMQ, QuaZIP, Socket Programming, QT, sqlite3, CouchDB, MySQL, Oracle, PostgreSQL, Cassandra, Apache Airflow, AWS EMR, AWS S3, AWS SES, Docker, K8s, Spinnaker, Aurora, Dynamo DB
Others	MS Excel, MS Word, MS PowerPoint, Operating System, Software Quality Management, VS 2015, Agile process, CMMI Level 5 process, Scrum, TDD (Agile), VMware Player, PSP, PDCA, RCA, JIRA, Stash, Jenkins, Perforce, Swarm, BOOST UT Framework, Google UT Framework, System Testing, Functional Testing, VS Code, IntelliJ.

PROFESSIONAL EXPERIENCE

Zendesk Inc

<https://www.zendesk.com/>
989 Market Street, San Francisco, CA 94103 USA

Zendesk Inc, Technical Lead

11th-Oct 2021 - Present

Zendesk is an American company headquartered in San Francisco, California. It provides software-as-a-service products related to customer support, sales, and other customer communications.

Project: ProxySQL

Responsibilities:

- Maintain/manage incoming database connections from different clients by multiplexing through ProxySQL layer with golang.
- Modify existing software to correct errors, allow it to adapt new hardware, or to improve its performance.
- Maintain high availability through K8s management coded in golang.
- Writing and managing smoke tests for ProxySQL usage.
- Maintain observability through Datadog.

Project: Znowflake

Responsibilities:

- Used DynamoDB and MySQL(Aurora) for generating ULIDs for large number of transactions across organization.
- Create and manage REST and GRPC servers for incoming requests and maintain microsecond level response time. Managed REST APIs and GRPC requests handling.
- Used AWS S3, EMR, Athena for Airflow data pipelines and querying results.
- Design, develop and modify software systems.
- Go integrations with GHA, Datadog, AWS sdk etc.

Develop and direct software system testing, refactoring and validation procedures, programming and documentation Determine system performance standards.
Consult with customers about software system design and performance.
Using Docker and K8s for the orchestration of different services across different pods.
System monitoring and observability through Datadog.

Project: Data-Lake and Analytics

Responsibilities:

Use AWS S3 as data-lake for all the streaming and batch data processing.
Implement kafka stream processing through continuous and periodic Airflow DAGs in python.
Use python operators for managing data processing orchestration.
Design, develop and modify software systems.
Use Java for batch and stream processing for big data (spark).
Onboarding new datasets seamlessly regularly.
Maintaining and managing AWS EMR jobs for data processing through spark.
Vulnerabilities managing across different data processing jobs.
Github automation for different processes.
Using Docker and K8s for the orchestration of different services across different pods.
System monitoring and observability through Datadog.

Environment: MAC, Linux, JIRA, VS, IntelliJ, GIT, AWS, Python, Golang, Java, Docker, REST APIs, GRPC, K8s, MySQL, DynamoDB, Aurora, Apache Spark, Apache Airflow, AWS S3, AWS EMR.

Tenerity Inc, (Earlier known as CxLoyalty/Affinion Group)

<https://www.tenerity.com/>

6 High Ridge Park, Stamford, CT 06905

Ph No: (972) 543-8515

Tenerity Inc, Senior Developer

16th-Dec 2019 - 8th Oct 2021

GoLang Application Developer/Technical Lead

Tenerity Inc. creates value for brands and their customers by partnering with clients to solve complex business issues and drive results. They design, build, personalize, and support customer experiences for some of the world's most successful brands. are a full-service customer engagement and loyalty company.

Project: Incentive Networks

The incentive networks team, spread across India and US works on backend APIs written in Golang, Python and Ruby to load and process transactions for different merchants and partners to find the loyalty rewards/rebates etc. for customers and enable them to have better engagement.

Responsibilities:

Write and manage transaction engine for rewards processing from scratch using Golang with REST API server as backend. Implemented authentication with go standard library for SHA256 and using salt mechanism.
For legacy code, implemented OAuth2 authentication.
Developed REST APIs to serve large data files as JSON and paginated responses.
Write and manage link(URL) checking for rewards/rebates related merchants and partners through web redirections in golang.
Modify existing software to correct errors, allow it to adapt new requirements, or to improve its performance.
Go integrations with aws/ses and k8s services.
Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.
Store, retrieve and manipulate data for analysis of system capabilities and requirements.
Design, develop and modify software systems, using scientific analysis and mathematical models to predict and measure outcome and consequences of design.
Develop and direct software system testing and validation procedures, programming, and documentation.

Determine system performance standards.

Environment: MAC, Windows, Linux, JIRA, VS, GIT, Bitbucket, Golang, Python, Docker, REST APIs, AWS.

CAPRUS IT Inc.

<http://caprusit.com/>

5750 Genesis Ct #135, Frisco, TX 75034

Ph No: (469) 920-8100

Project at Capital One (Contract), Plano, TX

30th-Oct 2018 - 13th-Dec 2019

GoLang Application Developer / Sr. Application Developer

Capital One Financial Corporation is a bank holding company specializing in credit cards, auto loans, banking and savings products headquartered in McLean, Virginia. Capital One is ranked 11th on the list of largest banks in the United States by assets. It is ranked 101st on the Fortune 500, 17th on Fortune's 100 Best Companies to work for list and conducts business in the United States, Canada, and the United Kingdom

Project: Sales & Origins / Loan Services

The S&O and LS teams works on different parts of data consolidation and data cleaning which is fed back to users and other stake holders to get a comprehensive view about the loan and account status.

Responsibilities:

- Implemented microservices in GoLang to facilitate an ETL framework.

- Used GraphQL to implement data schemas and query results.

- Incorporated and used Docker images for the proprietary plexus architecture.

- Used Mesos, Chronos for orchestration.

- Error/logs management using splunk.

Environment: MAC, JIRA, VS, GIT, GoLang, Beego, Python, Docker, REST APIs

Project: Tokenization and AutoID Image transfer

The tokenization handles sensitive customer data and tokenizes important information like SSN's so that no associates or anyone else can gather personal or financial information of customers. Auto ID image transfer is an autonomous system which handles image transfer from-to different databases for Auto Navigator application of Capital One in secured manner.

Responsibilities:

- Worked on JAVA and Spring Boot Framework to get Image transfer across DBs.

- Implemented microservices in Golang to do tokenization using turing.

- Tokenizing all PII information e.g. phone numbers, SSNs etc.

Environment: MAC, JIRA, GIT, GoLang, Java, Python, Docker, REST APIs, Mesos, Chronos, IntelliJ

Resolve Tech Solutions, Inc.

15851 North Dallas Parkway, Suite 1103, Addison, TX 75001

Project at CBRE (Contract), Dallas, TX

29th-May 2018 - 24th-Oct-2018

GoLang Application Developer

CBRE Group, Inc. is the largest commercial real estate services and investment firm in the world.[1] It is based in Los Angeles, California and operates more than 450 offices worldwide and has clients in more than 100 countries. Services provided by the company include facilities management services to occupiers of commercial real estate as well as property management, leasing, capital markets, appraisal, and brokerage services to owners of commercial real estate.

Project: Denali/Advisory & Transactions

Singh

The advisory and transactions team work on consolidation of data tables from different databases like PostgreSQL, MySQL, SQL etc. It serves as the front view for the researchers to work on a single database entity.

Responsibilities:

- Worked on Beego framework for developing a simple MVC application for DB query.
- Implemented microservices in GoLang.
- Used GraphQL to implement data schemas and query results.
- Incorporated and used Docker images for the proprietary lymbic architecture.
- Created/designed their MarketIQ backend framework.

Environment: MAC, JIRA, VS, GIT, GoLang, Beego, Python, Docker, REST APIs, Mesos, Chronos

Cloud 3 Technologies (Co-Op)

500 Lake Cook Road, Suite #350, Deerfield IL-60015

Chicago, IL

25th-September 2017 - 30th-November 2017

Software Design Engineer Internship

Cloud 3 Technologies uses exciting new techniques in software and solution development to build custom solutions to solve your unique challenges, without the custom costs. They use deep vertical experience in multiple industries, including Travel, Energy, Municipal & Public Safety, Transportation, and Telecommunications. They provide a powerful combination of domain-specific knowledge along with technical excellence to ensure your project has maximum impact at the lowest possible costs.

Project: EnforceUI (Exelon-TCPA)

The EnforceUI of Exelon-TCPA is a UI for managing customer details for usage of variety of Exelon services. This UI is built as a web app with backend in Java and front end in AngularJS.

Responsibilities:

- Worked on AngularJS and Java-based stack.
- Implemented process analysis to improve process.
- Facilitated the UI development using AngularJS using MVC.
- Incorporated dev-ops prerequisites for the team.

Environment: Linux, JIRA, STASH, GIT, SVN, AngularJS, Java, maven, tomcat, Kendo UI, REST APIs

Real Staffing (Co-Op)

222 West Adams, Suite 2450. Chicago, IL 60606

Project at Abbott (St. Jude Medical) (Contract), St. Paul, MN

22nd-May 2017 - 14th-July 2017

Software Design Engineer

Abbott is a global market leader in healthcare services and medical devices. Abbott has a broad range of branded generic pharmaceuticals, medical devices, diagnostics, and nutrition products. The company's in-vitro diagnostics business performs immunoassays and blood screening. Its medical tests and diagnostic instrument systems are used worldwide by hospitals, laboratories, blood banks, and physician offices to diagnose and monitor diseases such as HIV, hepatitis, cancer, heart failure and metabolic disorders, as well as assess other indicators of health.

Project: EnSite Velocity

The EnSite Velocity Cardiac mapping system is a suggested diagnostic tool in patients for whom electrophysiology studies have been indicated. It is helpful in displaying the position of catheters in the heart and do complex analysis.

Responsibilities:

- Was responsible for refactoring code in Qt/C++ for UI.
- Converted the current code base into Presenter first design pattern (MVP).

- Facilitated the test automation for UI.
- Incorporating TDD for UI.
- Used SOLID design principles.
- Implemented System and Integration test cases.
- Defined new use cases and fixed bugs.

Environment: Linux Red Hat, JIRA, STASH, GIT, C++, Swarm, QT/Linux, Perforce, Qt Creator, Ninja

Tektronix Engineering Development India

Salarpuria Premia, Survey No. 16 Kadubeesana Halli, Varthur Hobli Kaverppa Layout, Marathahalli - Sarjapur Outer Ring Rd, Bengaluru, Karnataka 560103, India

Tel.: [+91 80 3079 2600](tel:+918030792600)

Tektronix Inc., Bangalore, India

10th-June 2013 - 08th-July 2016

Technical Lead (June 2013 - July 2016)

Tektronix is a leading global Technology and IT enterprise offering services and products in Test and Measurement domain throughout the world. Tektronix, Inc. is an American company best known for manufacturing test and measurement devices such as oscilloscopes, logic analyzers, and video and mobile test protocol equipment. Worked on Tektronix's major application DPOJET and currently working on their upcoming future prospects. Released one product as "TekScope Anywhere" and currently working on their new product.

Project: TekScope Anywhere

TekScope Anywhere is a new Generation oscilloscope which is carved out of a common technology C++. It's a cross platform codebase running on both Linux and Windows. It's a very intuitive UI and makes it easier for the user to run and test measurements in Power/Jitter domain.

Responsibilities:

- Was responsible for developing algorithms in C++ for measurements and signal analysis.
- Implemented Jitter/Power algorithms and train team in technical quality.
- Implemented report infrastructure using HTML, CSS and JavaScript to generate reports in MHT and PDF.
- Incorporated TDD for measurements and wrote unit tests, integration tests and functional tests.
- Used SOLID design principles.
- Was responsible for implementing code quality and code standards.
- Maintained daily records of resolutions made by the team members and tracking the gap between the target achieved and the monthly target.
- Made sure that the customers are satisfied with the resolution of the defects.
- Took ownership of the issue faced by the customers and providing resolutions ASAP.

Environment: MS Visual Studio 2010/2012, JIRA, Stash, Git, C++, C# .NET, Quazip QT, Windows, Linux, HTML, CSS, JavaScript

Larsen & Toubro Technical Services(Earlier Integrated Engineering Services)

Worked as a Software Design Engineer II

KIADB Industrial Area Hebbal, Hootagalli, Mysuru, Karnataka 570018, India

Tel.: [+91 821 240 5572](tel:+918212405572)

Larsen & Toubro Limited, Integrated Engineering Services (L&T IES) 28th-February 2011 - 23rd-May 2013

Software Engineer (Embedded/Middleware)

Larsen & Toubro (L&T) is a technology-driven USD 12.8 billion company that infuses engineering with imagination. It offers a wide range of advanced solutions, services and products. L&T Integrated Engineering

Services (L&T IES) is a strategic business Unit of L&T Ltd. with expertise in various industry verticals' such as automotive, aerospace, industrial products, medical devices, consumer packed goods and science & technology. Their end-to-end service offerings include product design, analysis, prototyping & testing, embedded system design, manufacturing engineering, plant & construction engineering, asset information management & engineering process support using cutting-edge CAD/CAM/CAE technology in various domains.

Project: Lufkin Oil Well Manager (LWM)

Linux platform based application-oriented project in which the intent to develop some core applications, which could be run on ARM controller and eventually control the operation of oil pumping system. This is basically a pump controller that controls the pumping based on some inputs from the core application. These core applications include:

- DAL (Database Access Layer) which is an interface to the CouchDB database to store the data concerned with the pump system.
- EMC (External Communication Module) which is an interface between user and the external modules, which decides the parameters, that affects the pump system.
- SSI (Socket Server Interface) which serves as a channel for all the requests coming from a remote system in XML format to be served by other core applications.
- LEA (Logic Expression Application) which takes logic expressions to evaluate or set specific parameters or alarms of the pump system.
- MODBUS communication support through EMC.

All the core applications communicated through JSON protocol and the HMI communicates with SSI on XML protocol. Inter process communication was implemented using ZMQ library. Here, ZMQ's publisher/subscriber channel and request/reply channel was used.

Handled the code design issues and coded according to the best possible solution. By thorough review of the code and different use case scenarios, tried to find any design issues and resolved the same.

Coded development for the new requirement changes.

Performed application-level testing and executed different test cases for bugs or any other issues regarding the current design.

Extensively used GDB (GNU Debugger) debugging tool while performing testing activities on the project.

Provided detailed analysis of the requirement, decision making and informing resolution to the client concerned.

Made design adjustments in the code. With the change requests coming and to improve efficiency of the working code, sorted out the designs of applications.

Provided best possible resolution to the clients ensuring quality and customer satisfaction. Also, provided the most efficient and least time consuming methodologies to incorporate current and future formatted changes on the applications.

Provided feedback and suggestions to improve customer satisfaction based on the feedback received from the clients on issues reported.

Facilitated change management by taking monthly sessions to ensure that the team is updated with all the latest changes and procedures followed. Also, made sure that everyone in the team is on the same platform regarding that and there are no knowledge gaps within the team.

Logged all the information during the client calls. Served as the official logger of the client requirements during the client calls.

Gathered requirements in the initial stages of the project for all modules and then later on for specific module change requests.

Used SDLC for the applications development and followed the CMMI level 5 process to ensure quality of the released code.

Environment: LINUX in VMware, g++compiler, MS Word, MS Excel, MS PowerPoint, GDB, MS Outlook, Windows XP, Splice, MS Office

CERTIFICATIONS

"Machine Learning by Stanford University on Coursera" - Certificate earned on September 8, 2017

<https://www.coursera.org/account/accomplishments/certificate/S68WBNN3A7Q>

"Data Science Toolbox by John Hopkins University on Courser" - Certificate earned on November 26, 2017
<https://www.coursera.org/account/accomplishments/verify/KA3WWZNJDXRB>

Rest of the certifications mentioned at <https://techsingh.wixsite.com/website/certifications>